MobileHCI '25 Program

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Tue, 23 Sep

Session Content Gallery

Sort by: Time Title Shuffle

Papers

Contextra: Detecting Object Grasps With Low-Power Cameras and Sensor Fusion On the Wrist

Nathan DeVrio , Roger Boldu , Eric Whitmire , Wolf Kienzle

11:36 AM - 11:54 AM

Awards:

BEST PAPER

Abstract:

Knowing when a user picks up an object plays a vital role in many context-aware applications. For example, tracking water consumption, counting calories consumed, or reminding you to bring your keys are all context-centered scenarios involving picking up objects. In this project, we propose Contextra, a wrist-worn system that uses sensor fusion to...

Papers

Enhancing Smart Home User Experience: A Study of Everyday Objects for Smart Home Control

Michael Chamunorwa , Heiko Müller , Susanne Boll

11:54 AM - 12:12 PM

Abstract:

Current smart home technologies rely on touchscreens and voice assistants for interaction. These interfaces lack tactile engagement and fail to support users' daily routines and preferences, leading to poor user experiences (UX). Designing tangible user interfaces (TUIs) that align with user preferences can improve the status quo. This paper explores the...

Papers

Papers

in India

Amrit Singh , Vansh Bothra , Anirban Sen , Dipanjan Chakraborty

11:00 AM - 11:18 AM

Abstract:

Despite rapid mobile and Internet penetration in India, e-commerce adoption remains slow among emergent users. In this paper, we conduct a mixed-method study involving a two-phase user study and an analysis of app reviews posted on app stores. The findings reveal that emergent users often replicate offline shopping behaviours in digital contexts,...

Adversarial Use

Shaun Macdonald , Md Shafiqul Islam , Andras Bodrogai , James Finlow ,...

12:12 PM - 12:30 PM

Abstract:

Camera-equipped smartphones pose security risks to organisations by allowing intentional or accidental leaks of confidential on-screen information. We introduce SafeScreen, an Android camera app that detects and obfuscates screen content in real-time using deep-learning recognition for distant screens and Moiré pattern detection for close-up...

Papers

"Nuisance is Better Than Nothing?": Exploring How Pedestrians and Cyclists Perceive Automated E-Scooter Alerts in Shared Spaces

Hiruni Nuwanthika Kegalle , Danula Hettiachchi , Jeffrey Chan , Flora D. ...

11:18 AM - 11:36 AM

Abstract:

Electric scooters (e-scooters) offer flexible urban mobility but raise safety concerns in shared spaces. This study investigates how e-scooters can better communicate their presence to pedestrians and cyclists in shared active mobility environments. A focus group with e-scooter riders identified notifying others of arrival as a key challenge. To address this, w...

Papers

European Users' In-Depth Privacy Concerns with Smartphone Data Collection

Florian Bemmann , Maximiliane Windl , Tobias Knobloch , Sven Mayer

2:00 PM - 2:18 PM

Awards:

HONORABLE MENTION

Abstract:

Today's context-aware mobile phones allow developers to build intelligent and adaptive applications. The data demand induced by context awareness leads to decreased trust and increased privacy concerns. However, users' deeper reasons and real-world fears that underlie these concerns are not fully understood. We conducted an online survey (N=10...

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Mind Games: Exploring the Impact of Dark Patterns in Mixed Reality Scenarios

Luca-Maxim Meinhardt , Simon Demharter , Michael Rietzler , Mark Coll...

3:12 PM - 3:30 PM

Abstract:

Mixed Reality (MR) integrates virtual objects with the real world, offering potential but raising concerns about misuse through dark patterns. This study explored the effects of four dark patterns, adapted from prior research, and applied to MR across three targets: places, products, and people. In a two-factorial within-subject study with 74 participants, we...

Understanding User Prioritization and Comprehension of Smartphone Permissions

Manila Devaraja , Sameer Patil

2:18 PM - 2:36 PM

Abstract:

Smartphones allow users to control the sharing of their data with apps according to their privacy preferences. Yet, users struggle to enact their privacy preferences via the available permission settings. To understand whether these difficulties result from inaccurate understanding and/or suboptimal interface design of the permissions manager, we designed...

Papers

User Understanding of Privacy Permissions in Mobile Augmented Reality: Perceptions and Misconceptions

Viktorija Paneva , Verena Winterhalter , Franziska Hedwig Tania Augusti...

2:36 PM - 2:54 PM

Abstract:

Mobile Augmented Reality (AR) applications leverage various sensors to provide immersive user experiences. However, their reliance on diverse data sources introduces significant privacy challenges. This paper investigates user perceptions and understanding of privacy permissions in mobile AR apps through an analysis of existing applications and an onlin...

Papers

"An Ad Posing as Medical Advice": User Accounts of Dark UX in FemTech mHealth Apps

Ghada Alsebayel , Giovanni M Troiano , Johanna Gunawan , Herman Sak...

2:54 PM - 3:12 PM

Abstract:

FemTech is an emerging industry offering products, software, and services to support women's health and well-being. Within FemTech, mobile health applications (mHealth apps) are popular for managing menstruation, fertility, pregnancy, and menopause. Yet, these apps expose users to deceptive and misleading practices, which can be characterized...

Device Using a Distance Sensor and an IMU Sensor

Kai Miyashita , Takashi Amesaka , Shogo Hanayama , Takumi Yamamoto...

4:00 PM - 4:18 PM

Abstract:

Smart rings are used for contactless payment, smart lock operation, and health monitoring. For applications such as electronic payment and unlocking smart locks, the implementation of a user authentication system in smart rings is essential; however, some challenges remain. Fingerprint authentication is sensitive to fingertip conditions, while face...

Stylus Input Latency Compensation

kuangyu liu , Fangjie Hou , Xinyi Zhang , Jianlin Li , Zuobin Ning , Xuech...

4:54 PM - 5:12 PM

Abstract:

Input latency significantly deteriorates the users experience during touchscreen interactions, especially when they engage in precision tasks such as writing or drawing with a stylus. We address this issue by first decomposing it into two constituent tasks: stylus nib future trajectory prediction and predicted trajectory length optimization, facilitating a...

Papers

SwipeSense: Exploring the Feasibility of Back-of-Device Swipe Interaction Using Built-In IMU Sensors

Neel Shah , Benedict Leung , Mariana Shimabukuro , Ali Neshati

4:18 PM - 4:36 PM

Abstract:

The growing dimensions of smartphones have intensified the challenges associated with screen reachability. Back-of-device (BoD) interaction expands the range of reachability and offers a promising solution to mitigate screen occlusion while enhancing one-handed interactions. However, much of the existing research relies on incorporating additiona...

Papers

ThumbSwype: Thumb-to-Finger Gesture Based Text-Entry for Head Mounted Displays

Rishav Banerjee , Shariff AM Faleel , Omang Baheti , Khalad Hasan , Pou...

4:36 PM - 4:54 PM

Awards:

HONORABLE MENTION

Abstract:

Designing a comfortable, familiar, and efficient one-handed text entry method for Head-Mounted Displays (HMDs) remains a significant challenge. Existing midair typing systems induce fatigue, while novel techniques often demand extensive training or sacrifice input efficiency. Consequently, we introduce ThumbSwype, a novel thumb-to-finger text...

MobileHCI '25 Program

Demos

AI-Enhanced Landmark Recognition For Self-Guided Tour Application Using Large Language Models

Pronab Karmaker , Danai Korre , Muhammad Habib ur Rehman , masso...

6:00 PM - 6:05 PM

Abstract:

Artificial intelligence (AI), particularly Large Language Models (LLMs), has created opportunities to improve user experiences by enabling the development of more interactive applications in various implementation scenarios. This paper proposes a mobile application as a virtual self-guided tour, enabling landmark recognition and enhanced user...

Late-Breaking Work

An Evaluation Framework for Synthetic Touch Gestures – A Deeper Dive

Marvin Bachert , Marc Hesenius

6:50 PM - 6:55 PM

Abstract:

To enable automated testing of gesture-based applications, synthetic touch gestures must closely mimic real touch gestures. Although various generation methods exist, comparing them remains difficult due to the lack of standardized evaluation procedures. This work builds upon an existing initial evaluation framework that organizes relevant metrics into...

Late-Breaking Work

Are You With Me? Visualized Teleportation in Asymmetric VR-MR Collaboration

Amal Yassien , Tarek Soliman , Slim Abdennadher

6:55 PM - 7:00 PM

Abstract:

Locomotion is a key factor in virtual reality (VR) navigation but remains underexplored in asymmetric setups combining VR and mixed reality (MR). Our study investigates whether visualized teleportation can serve as a middle ground, balancing the social connection afforded by real walking with the efficiency of teleportation within asymmetric VR-MR setups. We...

Demos

Be Smart-Phone Zombie!: Guidance Display for Texting While Walking Using Striped Pattern and High-Speed Projection

Ryusuke Miyazaki , Shio Miyafuji , Hideki Koike

6:05 PM - 6:10 PM

Abstract:

Texting while walking, often referred to as a ``smartphone zombie'', has become a growing social issue that increases the risk of traffic accidents and pedestrian collisions. In this study, we propose a novel guidance display system that selectively presents visual guidance or warnings only to pedestrians using their smartphones while walking. The system...

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Demos

CamIO in the Browser: A Cross-Platform Audio Label Tool for Tactile Graphics

Dragan Ahmetovic , James Coughlan , Giorgio Dal Santo , Khadija Ezrour...

6:10 PM - 6:15 PM

Abstract:

For blind or low vision individuals, tactile graphics (TGs) provide essential spatial information but are restricted in the amount of data they can convey – especially semantic information, which is represented using the small amount of braille abbreviations that fit on the TG. Using computer vision, TGs can be enhanced by adding audio labels, in which audio...

Demos

CANViz: Low Cost Data Capture and Visualization of Automotive Signals

Kalani Murakami , Alexandre L. S. Filipowicz , David A. Shamma

6:15 PM - 6:20 PM

Abstract:

The automobile is a complex mobile multimedia system with an operating system that is mostly hidden and closed. For research, this creates two issues: collecting vehicle data for AI is arduous and rapidly building dashboard prototypes requires specialized expertise. We introduce CANViz to overcome these barriers and allow for data capture,...

Demos

ChemVR-AI: An Immersive Virtual Reality Laboratory with Pedagogical Agents

Rishi Jain , SWAYANSHU Rout , Gargi Srivastava

6:20 PM - 6:25 PM

Abstract:

Traditional hands-on laboratory education, while fundamental for skill development in the sciences, presents significant hurdles—including prohibitive costs, safety risks, and issues of accessibility. This paper presents ChemVR-AI, an immersive Virtual Reality (VR) laboratory designed to overcome these barriers in the context of chemistry....

Late-Breaking Work

Collaborating with LLMs Through a Voice and Graphical User Interface

Lenja Sorokin , Khanh Huynh , Malin Eiband , Lukas Stappen , Jeremy Di...

7:00 PM - 7:05 PM

Abstract:

The integration of Large Language Models (LLM) into systems that combine voice and graphical user interfaces causes a paradigm shift from designing for humans only to designing for collaborative interaction. However, the non-deterministic, natural language-based nature of LLMs also causes conflicts with the deterministic, structured GUI interaction. ...

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Late-Breaking Work

Comparative Evaluation of Visual Obfuscation Methods for Protecting Private Information in Screen Sharing

Rimi Makita , Mizuki Ishida , Kaori Ikematsu , Yuki Igarashi

7:05 PM - 7:10 PM

Abstract:

With the widespread use of online conferencing tools like Zoom on both PCs and mobile devices, there is a growing risk of unintentional exposure of private information through screen sharing. To address this issue, automated systems have been developed to detect and obscure private information on screens. However, the effectiveness of visual obfuscation...

Late-Breaking Work

Distributing In-Vehicle Interactions Across Devices with Insights into Drivers' Behavior, Preferences, and Usability

Laura-Bianca Bilius , Mihail Terenti , Radu-Daniel Vatavu , Jean Vanderdo...

7:10 PM - 7:15 PM

Abstract:

Connecting personal devices to the in-vehicle infotainment system has become mainstream in modern vehicles, contouring a distinctive context of use characterized by the user interface being distributed across multiple interactive systems, including those in the vehicle and the users' personal digital devices, likely involving different input and output...

Late-Breaking Work

DroDelivery: User-Centered Concepts for Intent-Based Communications of a Delivery Drone in Public Spaces

Shiva Nischal Lingam , Sebastiaan Martinus Petermeijer , Mohammad O...

7:15 PM - 7:20 PM

Abstract:

Drones will deliver packages in public spaces and interact with humans as recipients of the packages and as bystanders passing by. Clear communication of drone intentions is critical for reducing uncertainty and ensuring public safety. Limited research exists on interface designs addressing communication with diverse human roles. This user-centered...

Late-Breaking Work

E-static Sensors: Leveraging Triboelectricity for Enabling Gesture-based Interactions with Everyday Textiles

Ramyah Gowrishankar , Clayton Leite , Henry Mauranen , Yu Xiao

7:20 PM - 7:25 PM

Abstract:

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(TENG)-based sensing more accessible for e-textile-focused HCI research....

Late-Breaking Work

Four Replications of a User-Defined Gesture Study with Smart Rings

Jean Vanderdonckt , Radu-Daniel Vatavu

7:25 PM - 7:30 PM

Abstract:

Empirical findings in gesture-based interaction often stem from highly controlled experimental settings, which raises concerns about their generalizability. To explore how variations in such settings influence discoveries on user-defined gestures, we selected an end-user elicitation study involving smart rings that had been replicated at least once. By...

Late-Breaking Work

From Brick to Click: Comparing LEGO Building in Virtual Reality and the Physical World

Viktorija Paneva , Maximilian David , Jörg Müller

7:30 PM - 7:35 PM

Abstract:

We present an exploratory study comparing LEGO building experiences across three environments: the physical world, a Virtual Reality (VR) counterpart, and an enhanced VR setting with "superpowers" (e.g., infinite brick supply and first-person miniature perspective). Our aim is to investigate how creative physical activities can be translated into digital...

Demos

GUI-based Design System for Dynamic Light Performances with Portable Swarm Robots

Takumi Nishimura , Kunihiro Kato , Takashi Ohta

6:25 PM - 6:30 PM

Abstract:

We present a design system for dynamic light performances utilizing portable swarm robots. The system is lightweight and highly portable, enabling users to easily deploy it in various environments without spatial constraints. By assembling modular A3-sized metal plates, users can construct various performance stages, including those that use vertical...

Demos

Interactive Guidance for Self-Acquisition of Ultrasound Images by Hemophilic Patients

Dragan Ahmetovic , Claudio Bettini , Marco Colussi , Stefano Giacoia , Ro...

6:30 PM - 6:35 PM

Abstract:

Timely diagnosis of joint bleeds is crucial for preventing long-term damage in patients with hemophilia. However, access to specialized care and the operator-dependent nature of ultrasound imaging pose significant challenges for remote monitoring. We present GAJA (Guided self-Acquisition of Joint ultrAsound images), a mobile system that...

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Late-Breaking Work

LayEv: Advancing the Review-lution with Interactive Visualization of Systematic Evidence Synthesis

Luca Hilbrich , Alexander Kalus , Katrin Wolf

7:35 PM - 7:40 PM

Abstract:

With the rapidly growing number of articles published in the field of Human-Computer Interaction (HCI), it has become increasingly difficult to keep track of a given research area. Consequently, secondary research methods such as evidence synthesis are of growing importance. In this work, we transfer an evidence synthesis method from epidemiology to th...

Late-Breaking Work

Progressive Sentences: Combining the Benefits of Word and Sentence Learning

Nuwan Janaka , Shengdong Zhao , Ashwin Ram , Ruoxin Sun , Sherisse ...

7:40 PM - 7:45 PM

Abstract:

The rapid evolution of lightweight consumer augmented reality (AR) smart glasses (a.k.a. optical see-through head-mounted displays) offers novel opportunities for learning, particularly through their unique capability to deliver multimodal information in just-in-time, micro-learning scenarios. This research investigates how such devices can...

Demos

RehbeCa in Action! A Demo of Innovative Mobile Cervical Rehabilitation

Iosune Salinas-Bueno , Pau Martínez-Bueso , Maria Teresa Arbós-Bereng...

6:35 PM - 6:40 PM

Abstract:

Patients with cervical pain are often referred for rehabilitation treatment with a physiotherapist. This in-clinic treatment is usually combined with prescribed therapeutic exercises to perform at home. The dedication to the prescribed Home Exercise Program (HEP) is crucial to the effectiveness of the treatment and its beneficial long-term effects....

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